

ZHANG PENGPAI

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EDUCATION

City University of Hong Kong

MSc in Electronic Information Engineering GPA: 3.4/4.3

- **Core Courses:** Mobile Apps Design and Development, Machine Learning for Signal Processing Applications, Computer Vision, Complex Networks: Modeling, Dynamics & Control

China University of Geosciences

BEng in Computer Science and Technology GPA: 84/100

- **Core Courses:** Data Structures, Computer Networks, Operating Systems, Computer Systems, Database Systems, Algorithm Design and Analysis, Machine Learning

Hong Kong

Aug. 2025 – Feb. 2027

Wuhan, China

Sept. 2019 – Jun. 2023

PROFESSIONAL EXPERIENCE

Bank of China Software Development Center

Software Engineer

- Participated in the development of an internal banking management system, contributing to administrator login, API modification, data query, and exception-handling modules; supported account validation, multiple login methods, and mobile login integration with the bank's internal office platform.
- Wrote SQL scripts to identify and tag abnormal transactions and users, supporting manual review, risk reporting, and administrator-side follow-up workflows.
- Collaborated with developers, project leads, and client-side staff on requirement clarification, issue troubleshooting, deployment support, and user training.

Wuhan

Aug. 2023 – Apr. 2024

INTERNSHIP EXPERIENCE

Danaher China - Cytiva, Life Sciences

AI Application Development Intern

- Participated in an enterprise intelligence Agent system based on LangGraph, orchestrating multi-dimensional nodes for company identity resolution, financing transactions, financials, R&D pipeline, policy impact, capacity changes, and executive team analysis to generate structured company research reports.
- Contributed to company entity resolution and data-source routing by building company profiles with Chinese/English names, aliases, official websites, listing status, and stock symbols; designed a source-priority strategy using structured data first, official/regulatory sources second, and Web Search as fallback to reduce parent-subsidiary and listed-entity confusion.
- Supported facts-first output and dimension-level rule governance by structuring each node output into facts/meta/status/markdown; designed admission and noise-filtering rules for financing transactions, policy impact, and capacity changes to reduce low-quality web evidence and uncontrolled LLM generation.
- Improved Agent runtime stability through concurrency limits, timeout retries, context truncation, repeated-search detection, and node-level fallback for LLM, search, and web-fetch calls; in a 100-company batch test, improved end-to-end task success rate from about 85% to 96% and reduced repeated-run latency by about 45%.

Shanghai

Jul. 2026 – Sep. 2026

PROJECTS

QARA Agent: Azure OpenAI-based Regulatory Change Monitoring for Biopharma

AI Application Developer / RAG and Agent Engineering

- Built a regulatory-change monitoring assistant for biopharma and medical-device compliance, covering the end-to-end workflow of Bocha API search -> candidate cleaning -> local regulation matching -> structured report generation -> optional LLM impact analysis.
- Designed a regulation-candidate cleaning pipeline that normalizes Bocha raw JSON into RegulationCandidate, filters English pages, low-Chinese-ratio results, low-relevance pages, and duplicates before database/RAG ingestion, while preserving raw evidence and cleaned outputs for auditability.
- Implemented recursive local regulation scanning and incremental indexing for PDF, Word, TXT, and Markdown files; cached extracted text using path, file size, mtime, and sha256 fingerprints to avoid repeated full-corpus parsing.
- Built regulation retrieval and report generation with text chunks, keyword retrieval, Chroma vector indexing, and metadata boost to match new regulations with historical rules; integrated Azure OpenAI for difference analysis, risk-level judgment, and compliance recommendations.

May 2026 – Present

Redis-based Local Services Platform

Backend Developer

- Built a local services backend using Spring Boot, MyBatis-Plus, MySQL, Redis, and Redisson, covering user login, shop query, flash-sale ordering, and social interaction modules.
- Implemented Redis-based distributed session management with SMS-code login by replacing HttpSession with token + Redis Hash, and combined interceptors with ThreadLocal for authentication and session refresh.
- Designed a Redis cache layer using the Cache-Aside pattern, and handled cache penetration, hot-key breakdown, and consistency through null-value caching, mutex locks, logical expiration, and cache invalidation after database updates.
- Optimized the flash-sale workflow with Redis pre-deduction, Lua-based atomic validation, and Redis Stream asynchronous processing; combined Redisson locks, database checks, and Pending List recovery to guarantee one-order-per-user and reliable message handling.

Dec. 2025 – Mar. 2026

SKILLS

AI / LLM: LangChain, LangGraph, RAG, Agent, FAISS, BM25, Prompt Engineering, Vector Retrieval

Programming: Java, Python, SQL, JavaScript, HTML/CSS

Backend: Spring Boot, MyBatis-Plus, Redis, MySQL, RESTful API Design

Tools: Git, Docker, Nginx, Swagger/OpenAPI

Languages: Mandarin Chinese (Native), English (IELTS 7)